STUDENT PROJECT · SEVERE ASTHMA · SINGLE-CELL AI

# BiologicTraj

## How do asthma biologics reshape immune cells over time? Benchmarking PerturbGen and scGPT, read through LongOmicsTraj.

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**8**
patients with severe eosinophilic asthma

**3 visits**
before, 1 month, 6 months on biologic

**89,728**
PBMCs after QC, 24 samples

**4 + 4**
anti-IL-5 vs dupilumab (anti-IL-4Rα)

## One dataset, three lenses

**PBMC scRNA-seq**
0 → 1 → 6 months, per patient and cell type

**scGPT** · foundation model
Embeddings; in silico gene deletion

**PerturbGen** · trajectory-aware
Predicts 6-month state; perturbs T2 genes

**LongOmicsTraj** · trajectory topology
Labels every gene and cell type by its shape

**Benchmark + biology**
Which model gets endpoint and shape right

## Four questions

**AIM 1 · REPRESENTATION**
Do foundation-model embeddings separate before from after treatment, and drug class, better than PCA or scVI?

**AIM 2 · PREDICTION**
From a patient's baseline cells, can PerturbGen predict the 6-month state? Tested on held-out visits and held-out patients.

**AIM 3 · IN SILICO PERTURBATION**
Does knocking down *IL4R* or the IL-5 axis in baseline cells shift them the way real treatment did?

**AIM 4 · TOPOLOGY**
Which shapes of change are shared or drug-specific, and do model-predicted trajectories reproduce them?

## Three visits, nine shapes

Each step between visits is up, down or flat. Little changed by 1 month and most by 6, so delayed shapes should dominate.

| | | |
|---|---|---|
| up_up | up_flat | up_down |
| **flat_up** | flat_flat | **flat_down** |
| down_up | down_flat | down_down |

**Highlighted: delayed response, expected for NF-κB/AP-1 genes**

## Why it matters

- A first real-patient test of trajectory-aware perturbation models in asthma.
- Scores models on the **shape** of change, not just the endpoint.
- First single-cell application of LongOmicsTraj.
- No trajectory analysis of this cohort has been published.

**STATUS**
Planned. Results: [ADD RESULTS]

Data: Park et al., *Exp Mol Med* 2024 (PRISM, NCT05164939). Methods: Cui et al., *Nat Methods* 2024 (scGPT); PerturbGen, bioRxiv 2026; LongOmicsTraj, bioRxiv 2026.

**CODE**
`github.com/[USER]/BiologicTraj`